



Practice graded assignment: course project

🌟 AI Grading

After submitting your assignment and completing your required peer reviews, you'll receive an AI-generated grade based on the assignment rubrics. You'll then have the option to have your assignment reviewed by your peers instead.

Your data will be used in accordance with [Coursera's Privacy Notice](#).

Instructions My submission Peers to review Discussions

In this project, you will demonstrate your ability to perform the following:

- Create a key and use the Key Vault to secure the artifact.
- Deploy and manage Key Vault certificates, secrets, and keys to secure access to your applications.
- Configure Microsoft Graph permissions.
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- Configure Microsoft Graph permissions.

You will create two deliverables that document the steps and screenshots of activities related to securing applications. These deliverables are:

- Key Vault project solution
- Graph Explorer project solution

Your detailed tasks will include the following:

- Create a Key Vault
- Save a secret in Azure Key Vault
- Create a DevTest Lab
- Create a virtual machine in DevTest Lab and use the saved secret
- Complete the virtual machine deployment
- Create a key in an Azure Key Vault
- Use Microsoft Graph Explorer to run specific queries

At the end of this project, you will have the following deliverables for submission:

- *Key Vault project solution.docx*
- *Graph Explorer project solution.docx*

As part of this project, you will also review the project submissions of two (2) peers using the peer review rubric provided.

You should plan to complete this course project in about thirty (30) minutes. Also, allow up to one hour to complete the graded quiz that follows this project.

^ Step-By-Step Assignment Instructions

Refreshing your knowledge

Before starting on the project or while doing the project, you can refresh your knowledge by referring to the readings, videos, and associated exercise/solution documents in the below lessons:

- Week 1, Lesson 3: Manage Key Vault keys and secrets
- Week 3, Lesson 2: Microsoft Graph

Project scenario

In this project, you will go through the typical activities that an Azure security engineer would perform in an

organization.

Imagine you've recently joined a large organization as an Azure security engineer. You are responsible for acting on two problem statements. You will see the problem statements and the tasks to address the problem statements below:

Problem statement 1: More and more virtual machines are being created in the organization to support a new marketing solution. The security team just found out that developers are using non-compliant passwords. Passwords are also being shared across development and quality assurance teams. This practice is against the company policy. As a security engineer, you are concerned that this might lead to compliance issues during an audit. You decide to implement a more effective solution to use secrets during virtual machine creation.

Task 1: Store a secret in a Key Vault and use it during virtual machine creation.

Problem statement 2: The head office has selected your division to audit security best practices and compliance. As part of the audit, you must provide the information below:

- The number of guest users in your organization
- The top 10 alerts from the security center
- List of high-severity alerts
- List of all users in the Finance department
- List of devices associated with Azure Active Directory

Task 2: Run relevant queries using Microsoft Graph Explorer to provide the required information.

Setting up the system

This project consists of tasks you should complete on the Azure platform. Before you begin, you must have the following:

- Access to an Azure Active Directory instance as a Global Administrator
- An application, such as Snipping Tool, to capture screenshots
- An application, such as Microsoft Word, to create the two deliverables: *Key Vault project solution* and *Graph Explorer project solution*

Guidelines for the deliverables

We've provided two templates to help you document your work and build your deliverables. The templates provide starting points for the tasks you need to complete.

1. Select the template titled *Key Vault project solution template*.



Key Vault project solution template
DOCX File

2. Rename the downloaded template with a unique file name so you can easily find and upload it for submission later. For example, "Key Vault project solution_Your initials."

3. Select the template titled *Graph Explorer project solution template*.



Graph Explorer project solution template
DOCX File

4. Rename the downloaded template with a unique file name so you can easily find and upload it for submission later. For example, "Graph Explorer project solution_Your initials."

Guidelines for screenshots

1. Use an application, such as Snipping Tool, to capture screenshots.
2. Perform a step in the task and capture the corresponding screenshot.
3. Confirm that the screenshot displays the result of your activity. You should set your system display resolution to the highest resolution available.
4. You may also want to mask or remove personal information, such as sign in credentials, in the screenshot.
5. Copy and paste the screenshot below the appropriate step in your version of the relevant deliverable: *Key Vault project solution* or *Graph Explorer project solution*.

^ Completing project tasks

Creating a well-formatted document with steps and screenshots will help your peer effectively review your submission. Follow the instructions below to create your project deliverables.

Deliverable 1: Key Vault project solution

Task 1: Store a secret in a Key Vault and use it during virtual machine creation.

Activity 1: Create a Key Vault.

While performing this activity, document each step and the associated screenshot in your version of the *Key Vault project solution* document.

You can log in to the Azure portal and document the steps and screenshots. Remember to include the final screenshot that shows the deployment completion message.

The template provides examples of documenting the steps to make it easy for a peer to review your work. You will find the first two steps of this activity recorded in the template as a reference.

Create a Key Vault with the following specifications:

- Resource group name: **Ungraded-keyvault**
- Key Vault name: **secret-keyvault-new**
- Region: **East US**

Activity 2: Save a secret in Azure Key Vault.

While performing this activity, document each step and the associated screenshot in your version of the *Key Vault project solution* document.

You can use the Azure Key Vault that you created in the previous activity to perform this activity.

Create a secret with the following specifications:

- Upload options: **Manual**
- Name: **Password**

Remember to include the final screenshot that shows the secret named *Password* listed on the **Secrets** page.

Activity 3: Create a DevTest Lab.

While performing this activity, document each step and the associated screenshot in your version of the *Key Vault project solution* document.

Remember to include the final screenshot that shows the deployment completion message.

Create a DevTest Lab with the following specifications:

- Resource group: **Ungraded-keyvault**
- Lab name: **Dev-ungradedlab**
- Location: **East US**

Activity 4: Create a virtual machine in DevTest Lab.

While performing this activity, document each step and the associated screenshot in your version of the *Key Vault project solution* document.

Remember to include the screenshot that shows the use of the saved secret while creating the virtual machine.

Create a virtual machine with the following specifications:

- Virtual machine image: **Windows 11 Enterprise** (You can select other options as well)
- Virtual machine size: **B2s**
- Add **Chrome** as an artifact
- **Make this machine claimable: Yes**

Note: It might take about 30 minutes to display the password saved as a secret.

Activity 5: Complete the virtual machine deployment.

While performing this activity, document each step and the associated screenshot in your version of the *Key Vault project solution* document.

Remember to include the final screenshot that shows the virtual machine screen.

Task 2: Create a key in a Key Vault.

Activity: Create a key.

While performing this activity, document each step and the associated screenshot in your version of the *Key Vault project solution* document.

Create a key with the following specifications:

- Key Vault name: **Key-vaults-secret** (You can also choose another Key Vault you created.)
- Key name: **Key1**
- Remember to include the final screenshot that shows the successful creation of the key named *Key1*.

Deliverable 2: Graph Explorer project solution

Task: Run queries in Microsoft Graph Explorer

Note: You must grant Microsoft Graph Explorer the correct permissions using **Modify permissions** if you get errors when you execute the queries.

Activity 1: Run a query to view the number of guest users in your organization.

While performing this activity, document each step and the associated screenshot in your version of the *Graph Explorer project solution* document.

Remember to include the screenshot that shows the query result in **Response preview**.

Activity 2: Run a query to see the top 10 alerts from the security center.

While performing this activity, document each step and the associated screenshot in your version of the *Graph Explorer project solution* document.

Check whether you need to update the default query to list the top 10 alerts.

Remember to include the screenshot that shows the query result in **Response preview**.

Activity 3: Run a query to see alerts with high severity.

While performing this activity, document each step and the associated screenshot in your version of the *Graph Explorer project solution* document.

Remember to include the screenshot that shows the query result in **Response preview**.

Activity 4: Run a query to view all users in the Finance department.

While performing this activity, document each step and the associated screenshot in your version of the *Graph Explorer project solution* document.

You can use the demo data if you have not created a Finance department. You must access demo data without signing in.

Remember to include the screenshot that shows the query result in **Response preview**.

Activity 5: Run a query to view all the devices associated with Azure Active Directory.

While performing this activity, document each step and the associated screenshot in your version of the *Graph Explorer project solution* document.

Remember to include the screenshot that shows the query result in **Response preview**.

^ Finalizing your assignment

Post completion of all the tasks, you can finalize your deliverable. Ensure that you've added the steps and screenshots for all the tasks.

Remove any instruction sections or examples provided in the original deliverable template. Your final deliverable should contain only your work. You are then ready to upload your deliverable in the **My submission** tab.

^ Grading Criteria Overview

Twenty-five (25) points are allotted for this final course project. The points are distributed across the various tasks and activities that you have completed while developing the following deliverables:

- *Key Vault project solution* (15 pts)

- *Graph Explorer project solution* (10 pts)

Your grade will be based on the following distribution of points across tasks.

Deliverable 1: Key Vault project solution

Task 1: Store a secret in a Key Vault and use it during virtual machine creation. (12 pts)

Activity 1: Create a Key Vault. (3 pts)

Activity 2: Save a secret in Azure Key Vault. (3 pts)

Activity 3: Create a DevTest Lab. (2 pts)

Activity 4: Create a virtual machine in DevTest Lab. (2 pts)

Activity 5: Complete the virtual machine deployment. (2 pts)

Task 2: Create a key in a Key Vault. (3 pts)

Activity: Create a key. (3 pts)

Deliverable 2: Graph Explorer project solution

Task: Run queries in Microsoft Graph Explorer. (10 pts)

Activity 1: Run a query to view the number of guest users in your organization. (2 pts)

Activity 2: Run a query to see the top 10 alerts from the security center. (2 pts)

Activity 3: Run a query to see alerts with high severity. (2 pts)

Activity 4: Run a query to view all users in the Finance department. (2 pts)

Activity 5: Run a query to view all the devices associated with Azure Active Directory. (2 pts)

^ Preparing for project submission

Do not proceed to the **My submission** and **Peers to review** tabs until you've completed the following deliverables:

- *Key Vault project solution*
- *Graph Explorer project solution*

Review, proofread, and edit your deliverables. Ensure it's ready to submit for peer review.

To submit the deliverables, upload your documents to the Coursera platform following the prompts in the **My submission** and **Peers to review** tabs.

^ Peer review

Giving, receiving, and incorporating feedback is a crucial professional skill. After you submit your deliverable, as the next activity, you will have to review and provide feedback on two (2) peers' submissions. Similarly, your submission will be reviewed by a peer. We've shared a Rubric to ease the review process. You will be presented with questions about your peers' submissions. For example, did your peer document all the key evidence? If prompted, you may have to provide more in-depth, written feedback for some questions. Please provide constructive feedback to help your peers understand what they did well and what they can improve.

^ Compare your work (Discussion)

After submitting your deliverables and completing two (2) peer reviews, you can access the exemplars *Key Vault project solution exemplar* and *Graph Explorer project solution exemplar*. These exemplars are step-by-step guides for the project tasks. You can compare the exemplars with your submission. Then, reflect on your learning and share your assignment experience with peers in the discussion forum.

^ After completion of the project

When working on your Azure subscription, at the end of a project, it's good practice to assess whether you still require the resources you created. Running resources can cost you money. You can delete resources individually or delete the entire resource group containing the resources. Here's how you can do this.

Task: Clean up resources

Use the PowerShell command **Remove-AzResourceGroup** to delete the resource group.

The screenshot displays the Microsoft Azure portal interface. The top navigation bar shows the user is logged in as 'AZ500labuser5@nu500'. The main content area shows the 'Ungraded-keyvault' resource group. The 'Overview' tab is selected, showing a search bar and a list of resources. Below the portal, a PowerShell terminal window is open, showing the command `Remove-AzResourceGroup -Name "Ungraded-keyvault" -Force -AsJob` being executed. The terminal output shows a table with columns: Id, Name, PSJobTypeName, State, HasMoreData, Location, and Command. The first row shows a job running on 'localhost' with the command 'Remove-AzResourceGroup'.

Id	Name	PSJobTypeName	State	HasMoreData	Location	Command
3	Long Running O...	AzureLongRunni...	Running	True	localhost	Remove-AzResourceGroup

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